

A new fuzzy fractal control approach of non-linear dynamic systems: The case of controlling the COVID-19 pandemics

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ABSTRACT

This article is presenting a first attempt on a proposed fuzzy fractal control method for efficiently controlling nonlinear dynamic systems. The main goal is to combine the main advantages of fractal theoretical concepts and fuzzy logic theory for achieving efficient control of nonlinear dynamic systems. The concept coming from Fractal theory, known as the fractal dimension, can be utilized to measure the complexity of the dynamic behavior of a non-linear plant. On the other hand, fuzzy logic theory can be used to represent and capture the expert knowledge in controlling a plant. In addition, fuzzy logic enables to manage the uncertainty involved in the decision-making process for achieving efficient control of a non-linear plant. We illustrate the proposed fuzzy fractal control method with the current worldwide situation that requires achieving an efficient control of the COVID-19 pandemics.

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1. Introduction

Fuzzy logic has been well recognized as an efficient method to achieve control in non-linear plants and there have been many successful real-world cases, ranging from control in home appliances to controlling autonomous mobile robots or vehicles. Initially, type-1 fuzzy logic, proposed by Zadeh [1] was successfully used in fuzzy control, but later type-2 fuzzy logic also proposed by Zadeh [2] and later developed by Mendel [3] and others [4,5] was also used. It is now well-known that type-2 fuzzy systems can outperform type-1 in noisy, dynamically changing, and highly non-linear situations [6]. On the other hand, fractal theory studies the complex geometrical forms in nature, which are found to have in most cases self-similar structures that can be studied with the concept of a fractal [7]. In particular, the construct of the fractal dimension offers a way to estimate the complexity of geometrical objects in nature [8,9]. The main idea of this concept is having a way to measure the dimension with values different from the traditional integer values, meaning for example, that an object on the plane can have a fractal dimension between 1 and 2 depending on its geometrical complexity. This can be shown to be useful in classifying objects or in time series prediction, like in [10].

In recent times there has been a very quick spread of COVID-19 worldwide, spreading to many countries in different regions of the world. For the specific case of Europe, some countries like Bel-

gium, Italy, Poland Spain, England, France, and Germany were hit hard with the spread of COVID-19, with many confirmed cases and deaths [11–16]. Later, in the American continent, Canada, United States, Mexico and Brazil have also suffer with the COVID-19 pandemic [17–20]. As a consequence, it is of great relevance that strong research work has to be undertaken to model and understand all facets of the Pandemic [21–23].

Regarding related work on fuzzy control and chaos, there exist some interesting articles in the literature. There are some previous works on combining fractal theory with computational intelligence techniques for control in different applications, like in biochemical reactors [9,24] or in simulation of non-linear systems [25]. In addition, the work on cascade control for telerobotic systems serving space medicine shows an interesting approach for controlling a complex dynamic system [26]. Also, the work on stable fuzzy logic control of a general class of chaotic systems shows the relations of chaos theory with fuzzy control stability [27]. Finally, the work on filter-based adaptive control for nonlinear systems with unknown parameters in state and output equations proposes to manage uncertainty on the parameters of the dynamical systems that are under control [28]. In summary, there are works on fuzzy control and chaos theory and there are also some preliminary studies on fractal theory combined with neural networks and fuzzy logic, but these works are different to the approach proposed in this paper, as will be describe below.

Regarding related work on recent approaches to control the COVID-19 pandemic, there exist some recent interesting approaches based on optimal control. For example, we can find

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the work on analytic results for a modified Susceptible-Exposed-Infected-Recovered (SEIR) model and comparison of different intervention strategies for COVID-19 problem [29]. In addition, the interesting work on projections and fractional dynamics of COVID-19 with optimal control strategies [30]. Also, we can find the work on optimal control of mathematical modeling of the spread of the COVID-19 pandemic with highlighting the negative impact of quarantine on diabetic people with cost-effectiveness [31]. Another interesting approach is the work on predicting the dynamical behavior of COVID-19 epidemic and the effect of control strategies [32]. Finally, we can find the work on Sensitivity assessment and optimal economic evaluation of a new COVID-19 compartmental epidemic model with control interventions [33]. In summary, all these works are presenting control approaches based on mathematical optimal control theory, which are very interesting, but mostly dependent on the mathematical models of the epidemic. On the other hand, the proposed approach in this article is mainly based on artificial intelligence techniques, like fuzzy logic theory that enables to directly represent expert knowledge by fuzzy rules that form a fuzzy model. In this way, the fuzzy controller does not depend on the mathematical model, but rather on an intelligent approach based on previous data and expert knowledge. In addition, in this fuzzy model the fractal dimension is used to provide more information and improve results.

Based on the previous existing works, we propose in this article a hybrid fuzzy fractal approach for intelligent control in which the general architecture of a fuzzy controller is maintained, meaning the use of fuzzy systems formed by a set of fuzzy rules, but now the fractal dimension is utilized as input information to the controller to improve its performance. The main goal, is that the fractal dimension of the dynamic behavior of a non-linear plant can provide an idea of the complexity of the control actions required for achieving stabilization. We have to say that such an approach has not been proposed before in the literature and can be considered as the main contribution of this paper. Simulation results show encouraging behavior of the proposed fuzzy fractal approach for intelligent control of non-linear plants. In particular, we highlight the potential of the approach with the important and relevant problem of controlling the COVID-19 pandemics. The main advantage of the proposed fuzzy fractal approach is that expert knowledge can be expressed by the fuzzy rules and the fractal dimension (although a mathematical concept) is not too difficult to be used and allows measuring the complexity of the dynamics. However, as a disadvantage, the control method may not be applicable in cases of real-time control (like for robots in rapidly changing environments) because the fractal dimension (due to its numerical computation) may not be updated so quickly as to have a new value that will be reflected in the real-time control. Of course, this is not a problem for the COVID-19 control case, as we can update its value after several days to have a new value of the fractal dimension that can then influence the control of the pandemic.

The remaining of the paper is organized in the following way. Section 2 is presenting basic fractal theoretical concepts and Section 3 reviews the essential fuzzy logic theory concepts. Section 4 is presenting the proposed fuzzy fractal approach. Section 5 shows experimental results of the proposed approach for the case of controlling the COVID-19 pandemics. Section 6 summarizes the discussion of results. Finally, Section 7 puts forward some conclusions of the paper and envisions future works.

2. Basic concepts of fractal theory

Recently, there have been significant advancements in comprehending the complexity of an object with the utilization of fractal theoretical constructs [7]. One particular case, is time series in finance and economics, which have shown scaled properties sug-

gesting that they have a fractal structure [24]. Also, the fractal dimension has been applied in a plethora of areas, like in medicine, robotics and control [25,34]. The fractal dimension is expressed as follows:

$$d = \lim_{r \rightarrow 0} [\ln N(r)] / [\ln(1/r)] \quad (1)$$

where $N(r)$ represents the number of boxes needed to cover an object and r is a measure of the size of the box (a square on the plane). The fractal dimension numeric value can be estimated by counting the number of boxes needed to cover the object for different r sizes and then a statistical logarithmic regression is applied to obtain the d value (this is the well-known box counting algorithm). In other words, by counting the number of boxes for different r sizes and after that performing a logarithmic linear regression, we can find the box dimension of a geometrical object as follows:

$$\ln N(r) = \ln b - d \ln r \quad (2)$$

where d is the estimation of the fractal dimension, and the least squares method can be used to estimate this value.

The fractal dimension construct offers a powerful way to characterize and classify sets of objects. The main reason behind this statement is that the fractal dimension measures in some way the geometrical complexity of the objects. In particular, a time series can be classified based on the estimation of the numeric value of the fractal dimension. The reasoning behind this fractal classification method is that when the curve (formed with the time series) is smooth, then its fractal dimension will be close to a numeric value of one (considering the curve on the plane). On the other hand, when the curve is rougher, the fractal dimension will be higher and will tend to be closer to a value of two (as it will be covering more area on the plane).

3. Basic concepts of fuzzy logic

A fuzzy system can be used as a control scheme if an appropriate granulation of the input space can be performed. In this particular situation, the input space is defined by the time series of the relevant variables. The main goal is to be able to discriminate among different control actions by their properties using a set of fuzzy rules. We will assume, without losing generality, that the control problem is on bidimensional space. The data in this case is a time series that exhibits the behavior of the dynamic system (or plant). In this stage, we can begin by using fuzzy clustering [35,36] to agglomerate the data, and then constructing a set of fuzzy rules that will define a control method for the particular application.

Assuming that there are n Control Actions $CA_1, CA_2, \dots, CA_n$, and that fuzzy clustering can be applied to find n pairs (X_i, Y_i) $i=1, \dots, n$, which can then be assigned to the respective centers of the n clusters. Then the set of fuzzy rules can be stated in a straightforward fashion as follows:

$$\begin{aligned} \text{If } X \text{ is } x_1 \text{ and } Y \text{ is } y_1 \text{ then } U \text{ is } CA_1 \\ \text{If } X \text{ is } x_2 \text{ and } Y \text{ is } y_2 \text{ then } U \text{ is } CA_2 \\ \dots \quad \dots \quad \dots \\ \text{If } X \text{ is } x_n \text{ and } Y \text{ is } y_n \text{ then } U \text{ is } CA_n \end{aligned} \quad (3)$$

These fuzzy rules can be utilized for performing the control task by finding the corresponding values of the control variable U . In cases with higher dimensionality, this approach can be extended in a straightforward fashion. However, the critical issue is that there can be an exponential growth in the number of rules that has to be dealt for. For accomplishing the design of the fuzzy system in (3) we need to outline the membership functions for the fuzzy values of the X and Y linguistic variables, and estimate their proper parameter values. Fuzzy systems can be of Mamdani form, like in (3) or using equations as outputs, like in the Sugeno form [37,38].

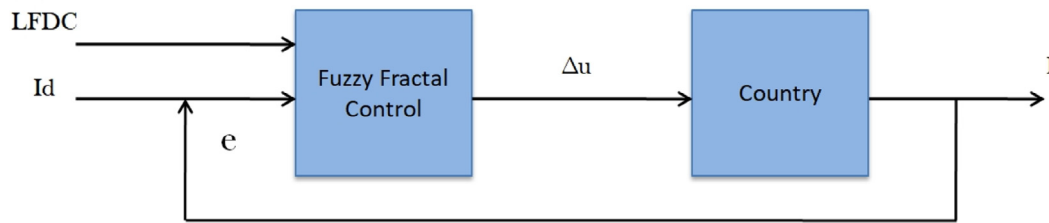


Fig. 1. General structure of the fuzzy fractal control scheme.

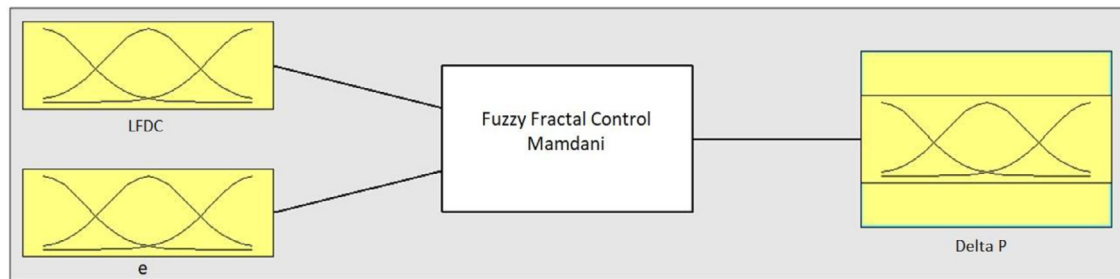


Fig. 2. Implementation of the fuzzy control scheme with two inputs and one output.

4. Proposed fuzzy fractal control method

In this Section, the fuzzy-fractal control method proposed in this article is outlined. The main idea is to combine the advantages of the fractal theoretical concepts and fuzzy logic theory, aiming at obtaining good controllers for non-linear dynamic systems. The fractal dimension can be utilized to estimate the complexity of the dynamic behavior of a non-linear plant. On the other hand, fuzzy logic theory can be used to represent and capture the expert knowledge in controlling a plant. One of the advantages of fuzzy control is that a mathematical model is not needed for deriving the fuzzy controller, although the relevant variables used are based on existing mathematical models for COVID-19, like the ones found in [39,40,41]. In Fig. 1 we illustrate the general structure of the control scheme, in this case for controlling the COVID-19 pandemic of a particular country. The observable variable is the number of Infected people I , and we propose a desired value I_d , to calculate the error e as one of the inputs of the controller. The other input is the linear fractal dimension of the country (LFDC) that measures the complexity of the dynamics of the plant. The output of the fuzzy fractal control is Δu , which represents the change in the control action that is in this case is the change in parameter p , representing the fraction of persons receiving a certain type of treatment (like the vaccine).

In this case, based on expert knowledge about control and also considering the knowledge about the fractal dimension concept, we can establish the following fuzzy rules for control:

- IF LFDC is low and e is low THEN Δu is low
- IF LFDC is high and e is low THEN Δu is medium
- IF LFDC is high and e is medium THEN Δu is medium
- IF LFDC is high and e is high THEN Δu is high

The knowledge comprised in these fuzzy rules can be explained as follows. The first rule states that if the fractal dimension is low and the error is low, then the change in treatment should be low, as we are near stabilizing the dynamics to an equilibrium state. Then we have fuzzy rules for when fractal dimension is high, but the error is low or medium, in which case the change in the treatment is considered to be medium, and finally if the fractal dimension is high and the error is also high the change in treatment should be high, as we are far from stabilizing the pandemics.

We illustrate in Fig. 2 the actual software implementation of the structure of the fuzzy control scheme, illustrating the use of

the two inputs and one output of the fuzzy system. In this Fig., we have two inputs, which are LFDC and e , and one output, which is the Δu that in Fig. 2 is shown as Delta P. In the Block label as Fuzzy Fractal Control is where the fuzzy model consisting of fuzzy rules is contained.

We show in Fig. 3 the implementation of the fuzzy rules in the software system. These fuzzy rules can be interpreted as providing the corresponding control actions (outputs) for each possible situation, which are indicated by the conditions of the fuzzy rules. For example, in Rule 4 we are expressing the knowledge that when the fractal dimension is high (complex dynamic behavior of the time series) and the error is high (we are far from the desired value of I), then the control action should provide a high increment to the p value (percentage of persons receiving a treatment for the illness). The other rules can be interpreted in a similar fashion.

We illustrate in Fig. 4 the three membership functions of the output linguistic variable of the change in the p value. The range of values of the increment p was found to be in the range between 0 and 0.2, and was granulated to be: low, medium and high. The actual result is obtained by calculating a weighted average (called defuzzification) over the resulting membership function or functions (for overlapping cases). In this case, we show in Fig. 4 the three Gaussian membership functions that are utilized for the corresponding values of low, medium and high.

We also illustrate in Fig. 5 the two membership functions that were used for the input variable of the linear fractal dimension for confirmed cases (LFDC). The range of values for the fractal dimension (for this application) was found to be between 1 and 1.65 and we only needed to use two fuzzy values of low and high for the dimensions. The membership functions are also Gaussian for the two linguistic values.

We also illustrate in Fig. 6 the three membership functions that were used for the input variable of the error (e). The range of values for the error (for this application) was found to be between 0 and 0.18 and we needed to use three fuzzy values of low, medium and high for the errors. The membership functions are also Gaussian for the three linguistic values.

Finally, we show in Fig. 7 the nonlinear surface of the fuzzy fractal model representing the form in which the variables are related. The nonlinear surface is a representation of the complete fuzzy model, as it is graphical illustration of the three variables (two inputs and one output) that forms a three-dimensional sur-

1. If (LFDC is Low) and (e is Low) then (Delta P is Low) (1)
2. If (LFDC is High) and (e is Low) then (Delta P is Medium) (1)
3. If (LFDC is High) and (e is Medium) then (Delta P is Medium) (1)
4. If (LFDC is High) and (e is High) then (Delta P is High) (1)

Fig. 3. Implementation of the fuzzy rules of the fuzzy fractal model for control.

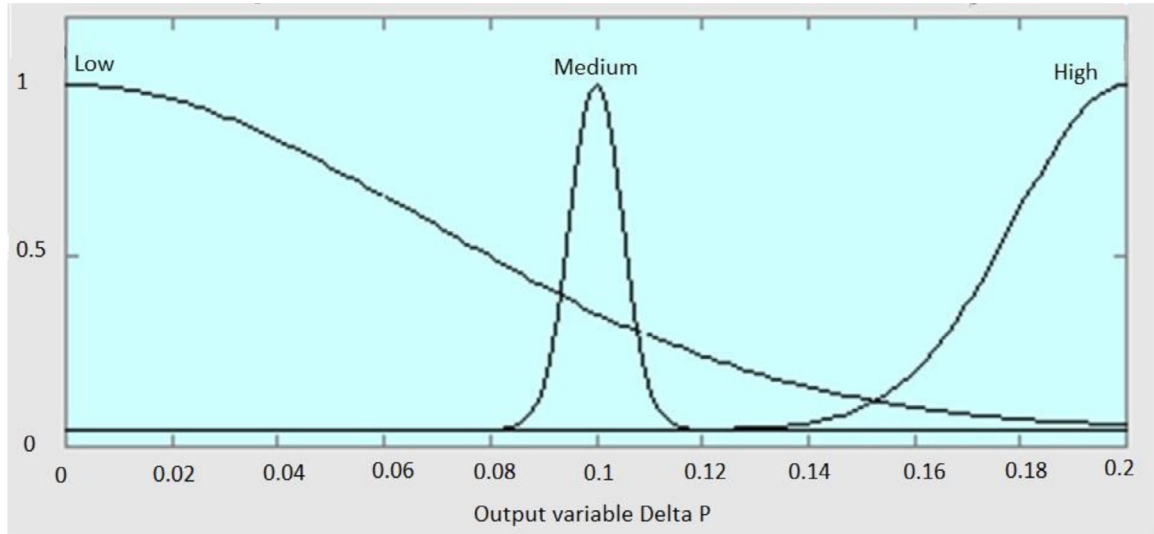


Fig. 4. Membership functions of the output variable of the fuzzy system.

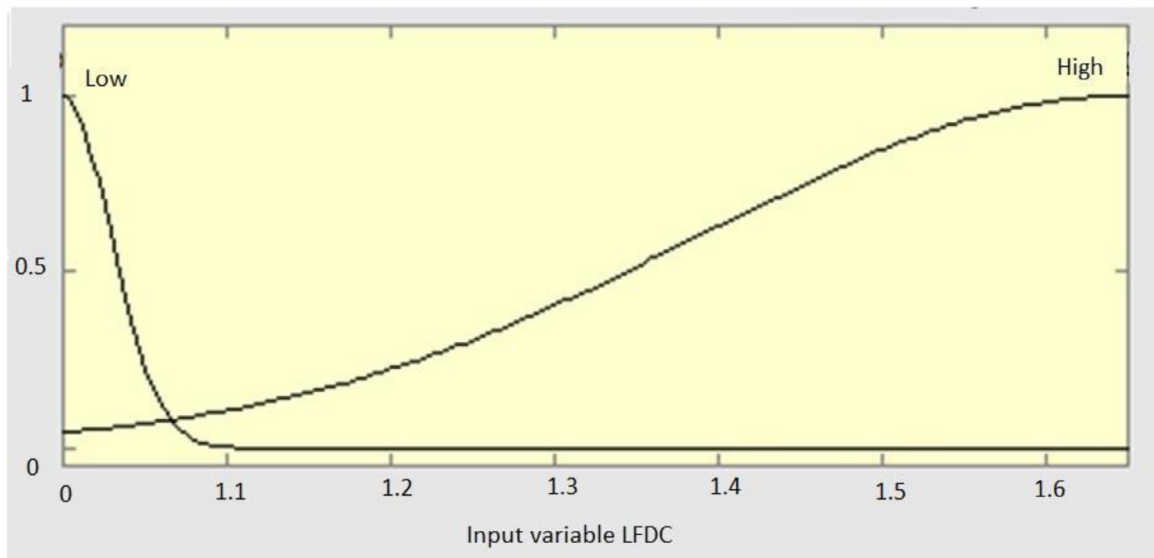


Fig. 5. Membership functions of the LFDC input variable of the fuzzy system.

face. In this case, the surface shows how the fuzzy model is relating the two inputs with the output. Basically, the fuzzy model is giving us a surface that we could also obtain with a mathematical model, but with a fuzzy system is easier to implement using fuzzy rules.

This fuzzy model for control uses as input the real data of the COVID-19 time series (data coming from Humanitarian Data Exchange (HDX), which is a publicly available dataset on COVID-19 [11] to calculate the control p . On the other hand, if we had used a traditional mathematical control method, then the COVID-19 model would be directly needed (requiring for example a numerical method to integrate differential equations, like the Runge-Kutta method, making the controller slower because of computa-

tion required). However, in our case with the proposed fuzzy fractal control approach we are directly using the real data as input, and then the outputs of the controllers are used to update the previous values in the time series. Finally, this time series produced by the controller is compared with the real data to measure the effect of the controller.

5. Simulation results

The experiments to test the method using the fractal dimension and fuzzy logic for control are presented in this section. First, the fractal analysis was performed according to the quantity of COVID-19 cases to find the numeric values of the fractal dimensions for the countries based on their historical data. The database

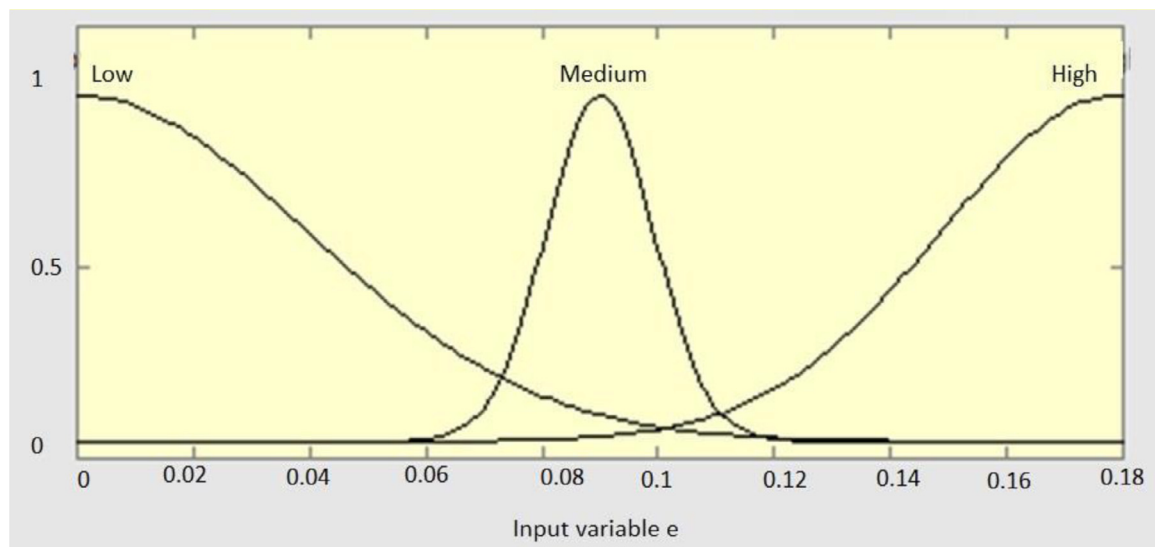


Fig. 6. Membership functions of error e input variable of the fuzzy system.

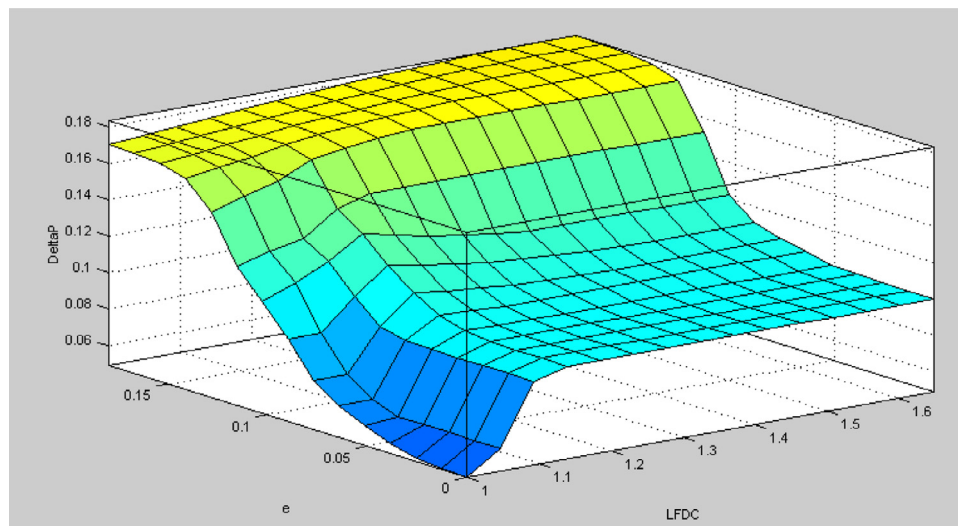


Fig. 7. Nonlinear surface of the fuzzy fractal fuzzy system for control.

that was utilized for performing the experiments was extracted from the Humanitarian Data Exchange (HDX) [11]. This database is composed of COVID-19 data from all the countries from January 22, 2020 to January 20, 2021. The consulted dataset for achieving the set of experiments and results presented in this section is the following one: `time_series_covid19_confirmed_global`. The data consists of the confirmed cases of patients from the countries. In Fig. 8 the plot illustrates the general trend in the time series data for Belgium, clearly showing the COVID-19 Confirmed cases for the time period of 22-01-2020 to 20-01-2021. In Fig. 9 a similar graphical representation for Brazil is illustrated. Fig. 10 similarly shows the confirmed cases for India. Fig. 11 also illustrates the plot of confirmed cases for Italy.

In Fig. 12 we show the confirmed cases for Japan, in Fig. 13 for Mexico and in Fig. 14 for USA for the same period of time (January 22 of 2020 to January 20 of 2021).

We also show in Fig. 15 the plots of the confirmed cases for the seven countries involved in this study. The countries are Belgium, Brazil, India, Italy, Japan, Mexico, and the United States of America (USA).

Based on the previous plots of the confirmed cases, we are able to find the numeric values of the fractal dimensions of the countries for the previously indicated time period of one year. In Table 1 we show the numeric values of the fractal dimensions of the countries, along with their corresponding fuzzy values (of the membership functions presented in Section 5) that are used in the fuzzy system

The next step was to use the fuzzy system shown in Fig. 2, using as input the fractal dimensions of the countries and the error in control e defined by:

$$e = I_d - I \quad (4)$$

where I_d is a desired valued of confirmed cases that is set according to each country to a number that we desire to achieve, and I represent to the actual value. For the experiments we did consider the last 10 days of the one-year period mentioned before, and we show in the following Figs. 16 to 22 the impact of applying the proposed fuzzy fractal approach for controlling the pandemic in the seven countries considered for this experimental work. In the Figs we only show the last 30 days (from December 22 of 2020 to January 20 of 2021) and the control approach (shown in blue)

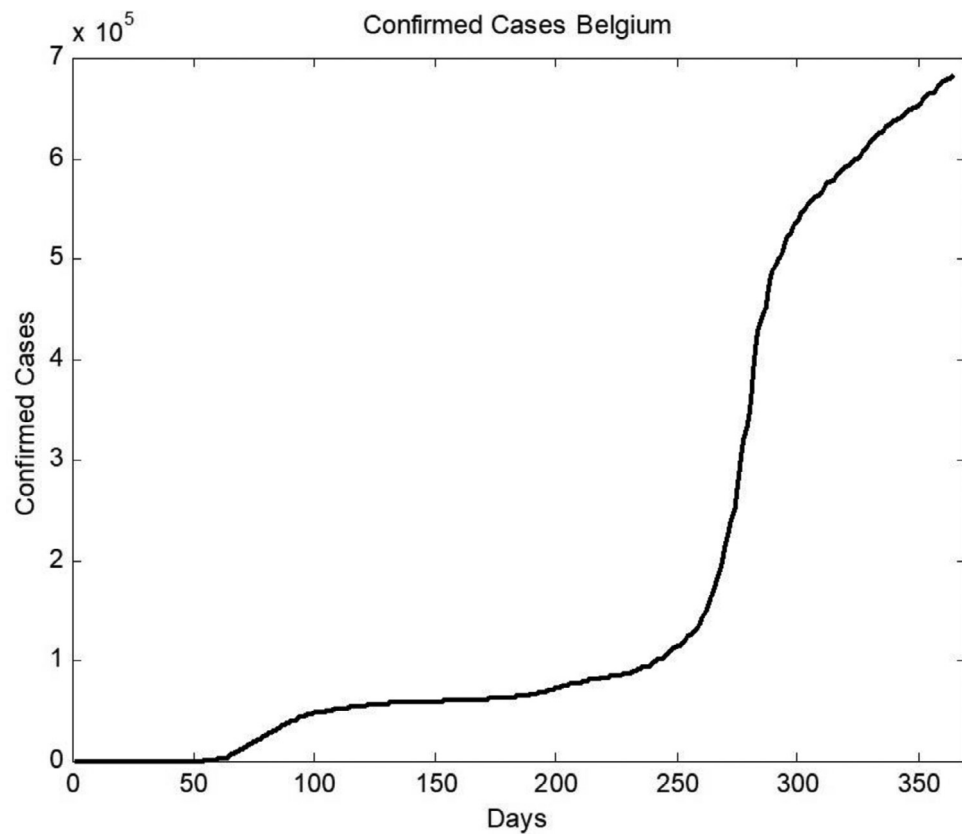


Fig. 8. Plot of confirmed cases for Belgium from 22-01-2020 to 20-01-2021.

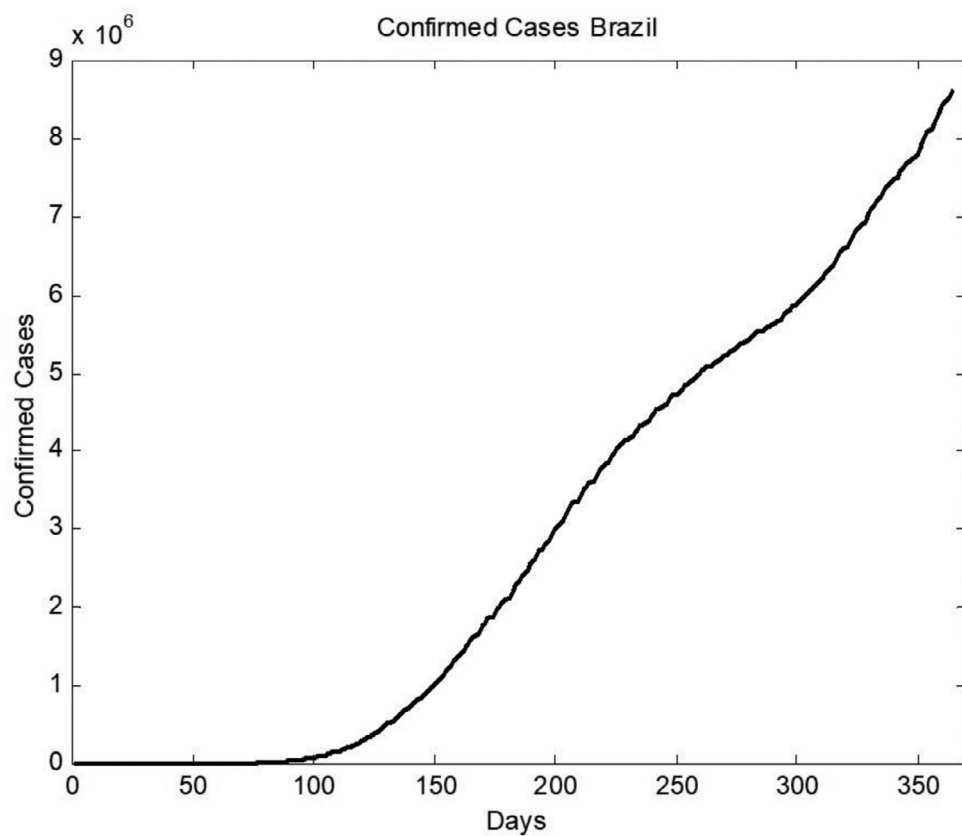


Fig. 9. Plot of confirmed cases for Brazil from 22-01-2020 to 20-01-2021.

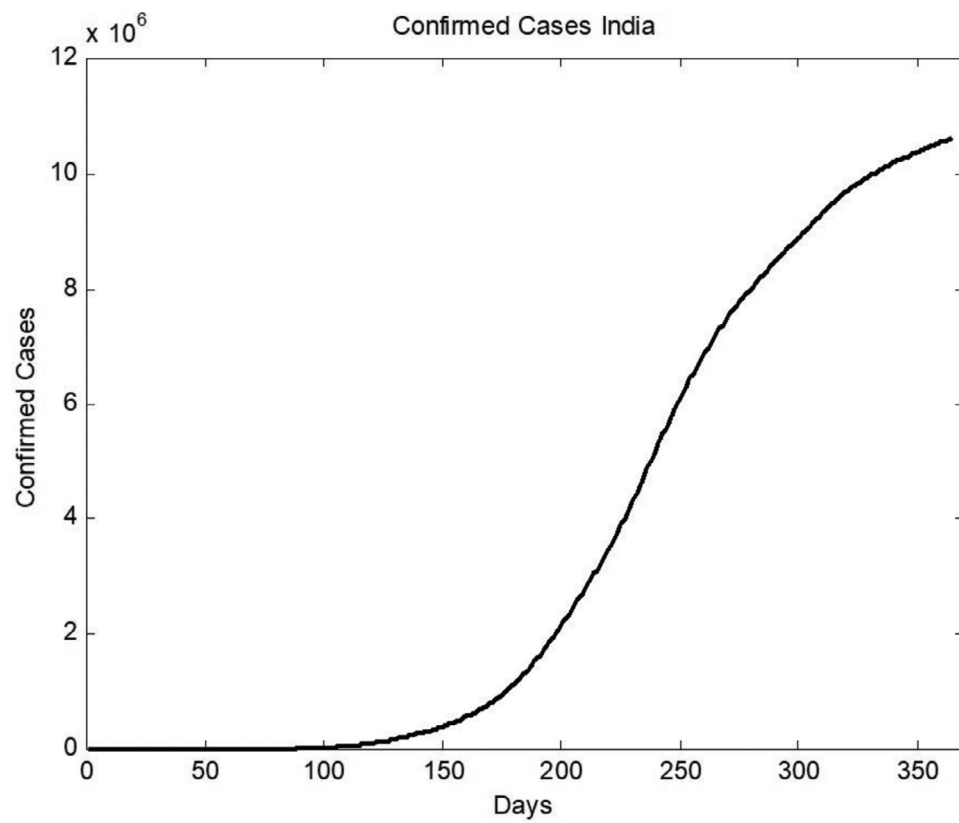


Fig. 10. Plot of confirmed cases for India from 22-01-2020 to 20-01-2021.

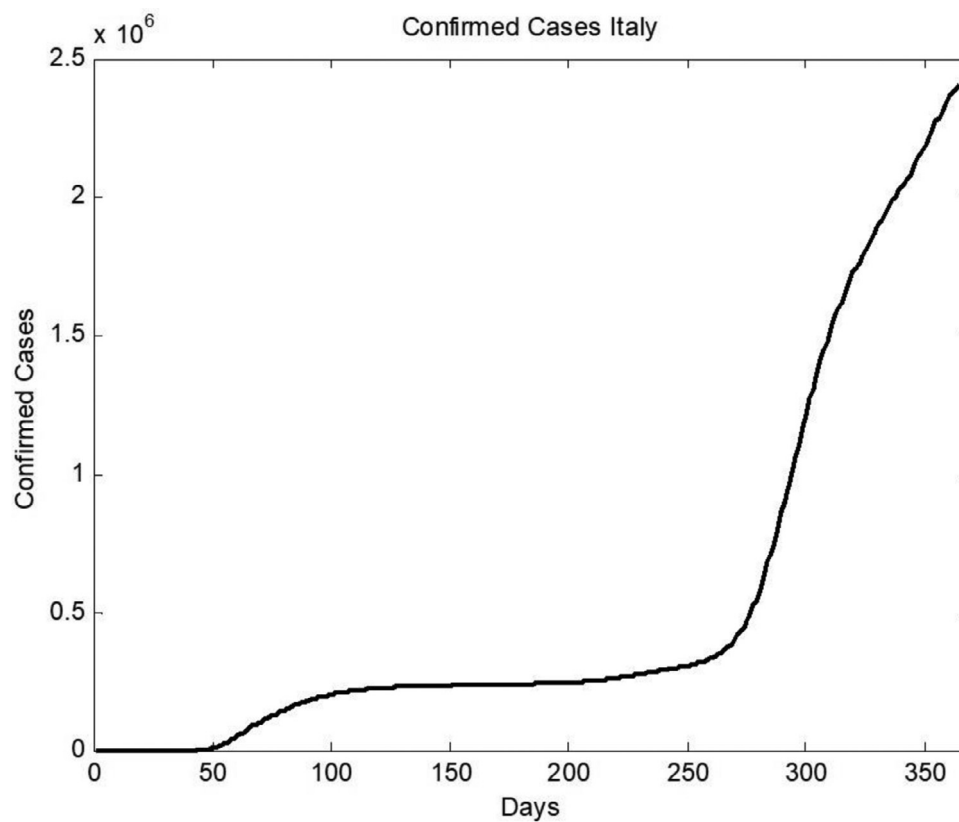


Fig. 11. Plot of confirmed cases for Italy from 22-01-2020 to 20-01-2021.

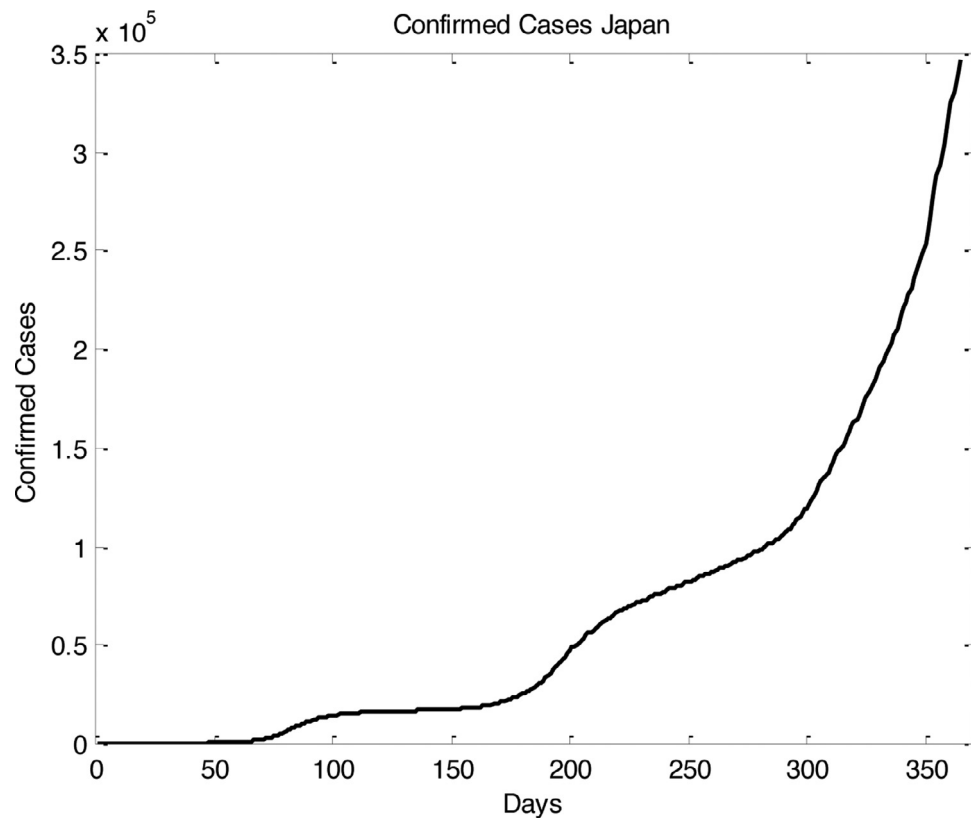


Fig. 12. Plot of confirmed cases for Japan from 22-01-2020 to 20-01-2021.

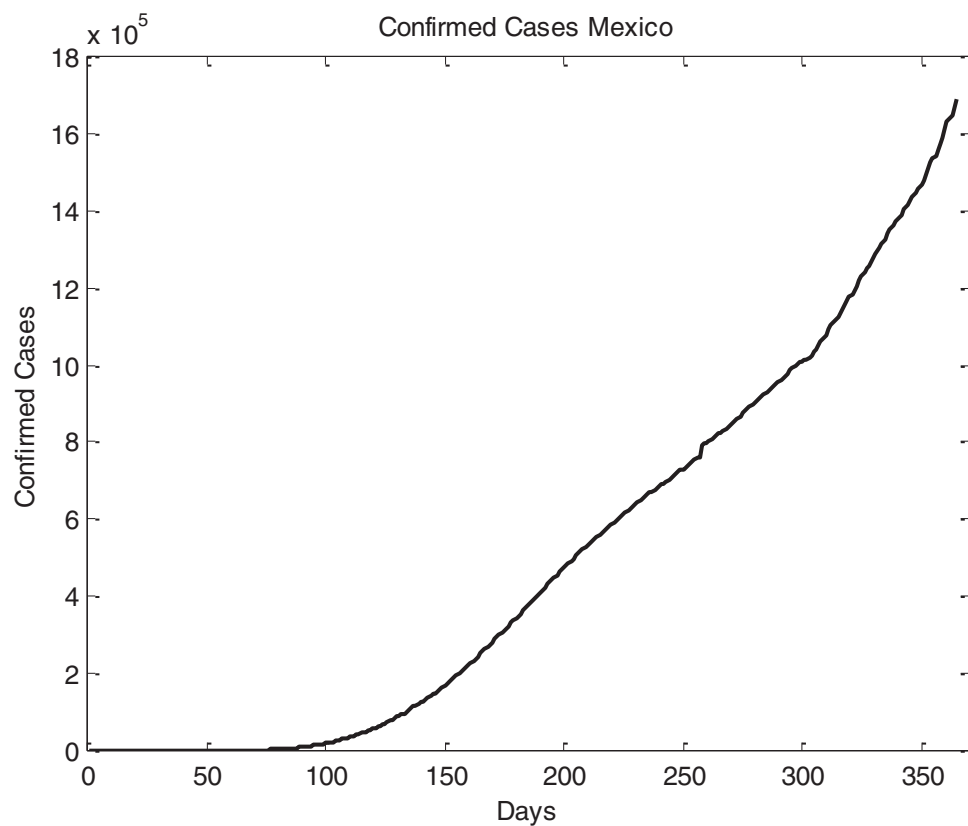


Fig. 13. Plot of confirmed cases for Mexico from 22-01-2020 to 20-01-2021.

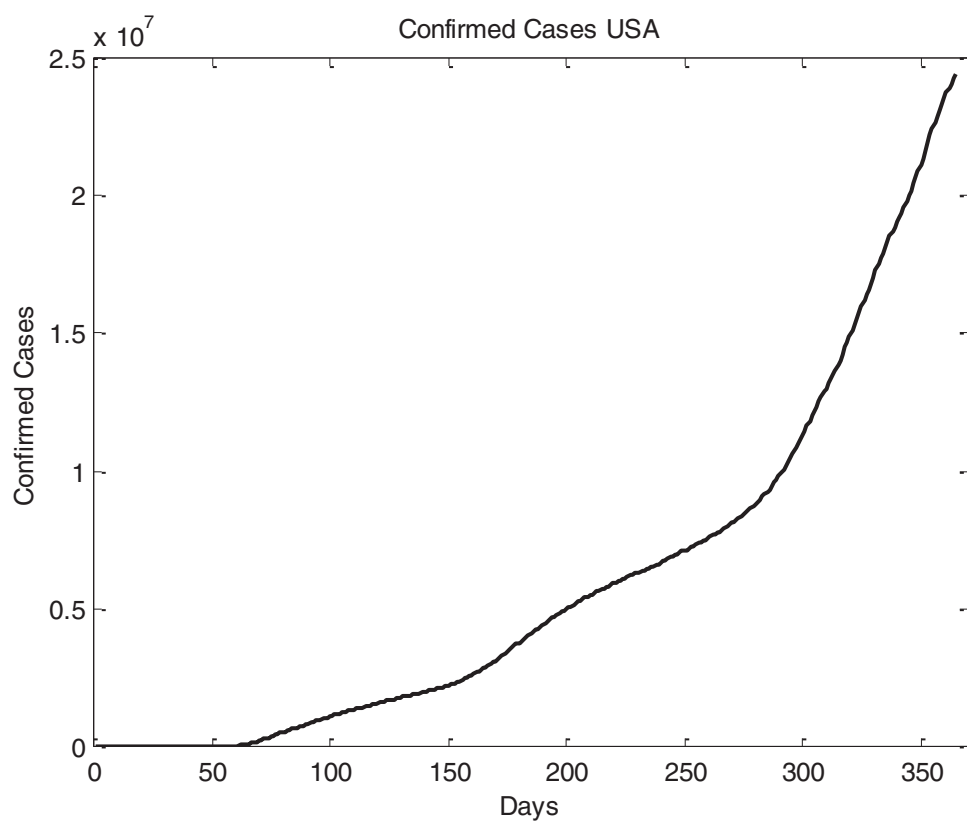


Fig. 14. Plot of confirmed cases for USA from 22-01-2020 to 20-01-2021.

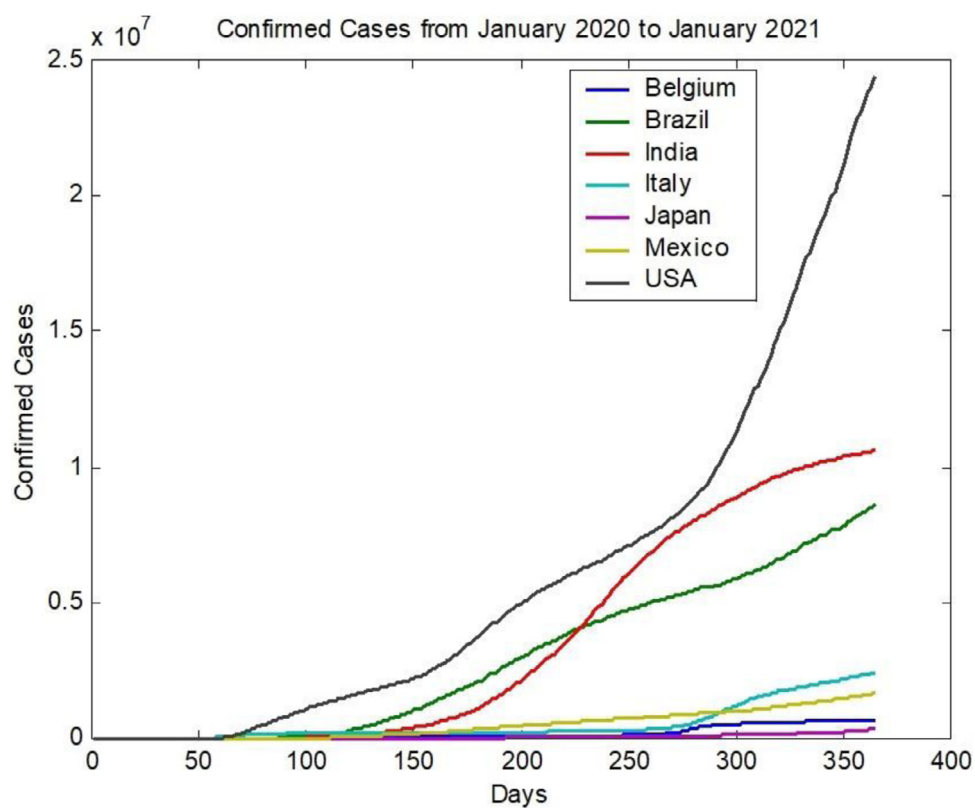


Fig. 15. Plot of confirmed cases for the seven countries from 22-01-2020 to 20-01-2021.

Table 1
Numerical fractal dimensions and corresponding fuzzy values.

Country	Fractal dimension LFDC	Fuzzy set with highest value of membership	Fuzzy set with lowest value of membership
Belgium	1.192	High (0.26)	Low (0.01)
Brazil	1.084	High (0.15)	Low (0.10)
India	1.18	High (0.24)	Low (0.01)
Italy	1.196	High (0.26)	Low (0.01)
Japan	1.195	High (0.26)	Low (0.01)
Mexico	1.185	High (0.25)	Low (0.01)
USA	1.182	High (0.24)	Low (0.01)

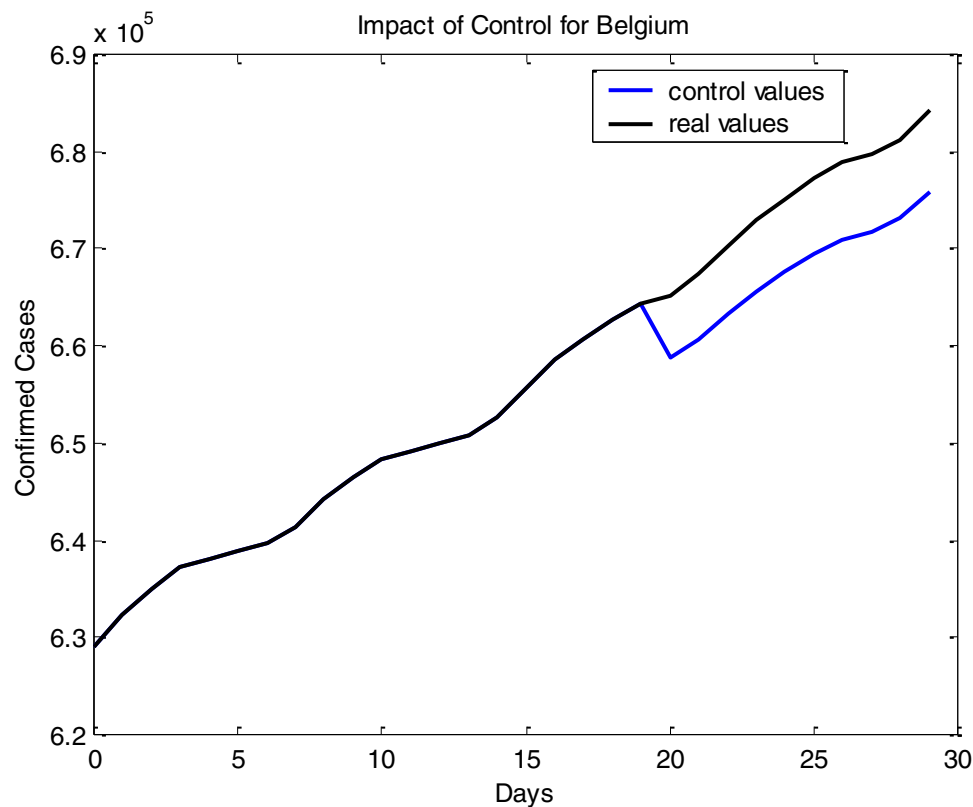


Fig. 16. Plot of real confirmed cases (black) and cases under the impact of the controller (blue) for Belgium.

clearly illustrates that the control method is achieving the goal of reducing the confirmed cases for all the countries. Of course, the impact of the controller is not the same for all countries, as in some of them the impact appears to be higher. For example, the effect of the controller on Belgium, India, Mexico and USA appears to be higher than in other countries.

We illustrate in Fig. 16 the impact of the fuzzy fractal control approach, where the curve in black is representing the real number of confirmed cases and the curve in blue shows the impact if the proposed approach had been used in the last 10 days of this period. The curve in blue clearly indicates that the number of cases could have been reduced by using the proposed approach. In Fig. 17 we are showing the same experiment for Brazil, where the effect of the controller, although noticeable, appears to be lower. In Fig. 18 we illustrate the results of the fuzzy fractal controller applied to the case of India, which shows a good impact in reducing cases. We also show in Fig. 19 the results of the controller for Italy, although in this case the impact is not as big, but there is a reduction on the number of cases. In addition, we illustrate the impact of the controller for Japan in Fig. 20 and in this case, we appreciate a good effect of the control approach. Finally, we are presenting in Figs. 21 and 22 the impact of the fuzzy fractal approach for Mexico and USA, respectively.

6. Discussion of results

In summary, we can state that based on the previous results, the proposed fuzzy fractal approach performs in a good way, as it is able to reduce the number of COVID-19 cases for the countries considered in the experiments. The impact is not the same on all countries, but this was expected as their dynamics are different. In considering similar works dealing with COVID-19 control [29–33], most of these papers are presenting control approaches based on mathematical optimal control theory, which are very interesting, but mostly dependent on having very good mathematical models of the epidemic and then being able to mathematically derive equations for control. On the other hand, the proposed approach in this article is mainly based on artificial intelligence techniques, like fuzzy logic theory that enables to directly represent expert knowledge on control by using fuzzy rules that form a fuzzy model. In this way, the fuzzy controller does not depend on the mathematical model, but rather on an intelligent approach based on previous data (that is collected daily) and knowledge from human experts on the problem. In addition, in this fuzzy model the fractal dimension is used to provide more information about complexity of the dynamics of COVID-19 data and in this way help to improve results. Finally, we have to say that in this work the design of the

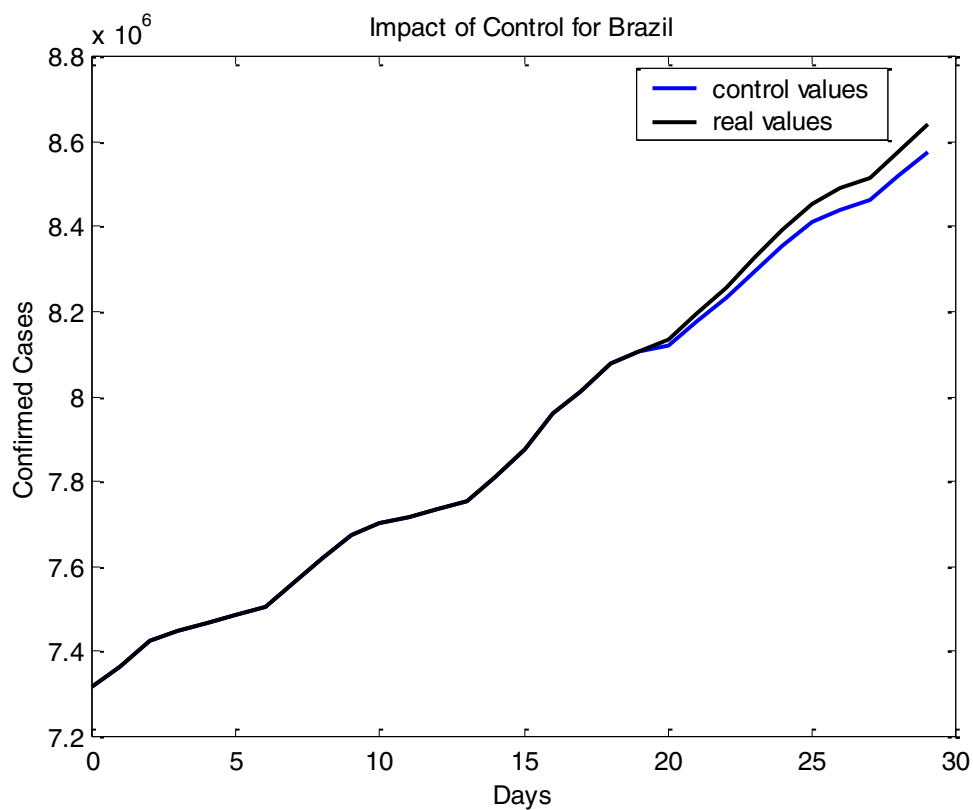


Fig. 17. Plot of real confirmed cases (black) and cases under the impact of the controller (blue) for Brazil.

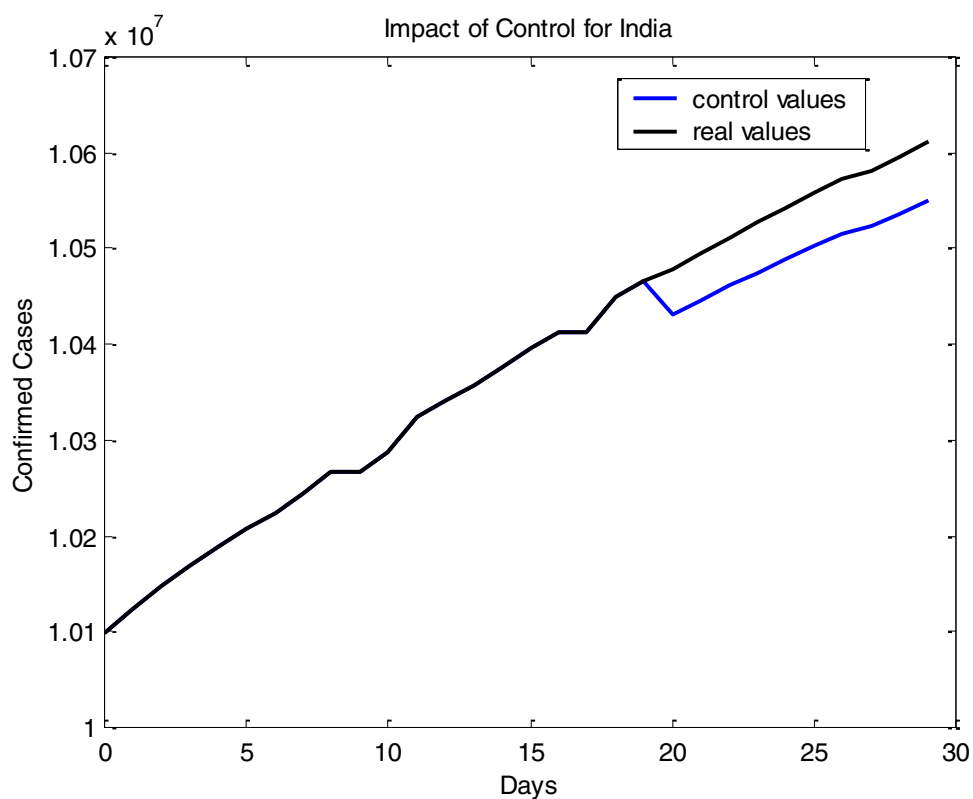


Fig. 18. Plot of real confirmed cases (black) and cases under the impact of the controller (blue) for India.

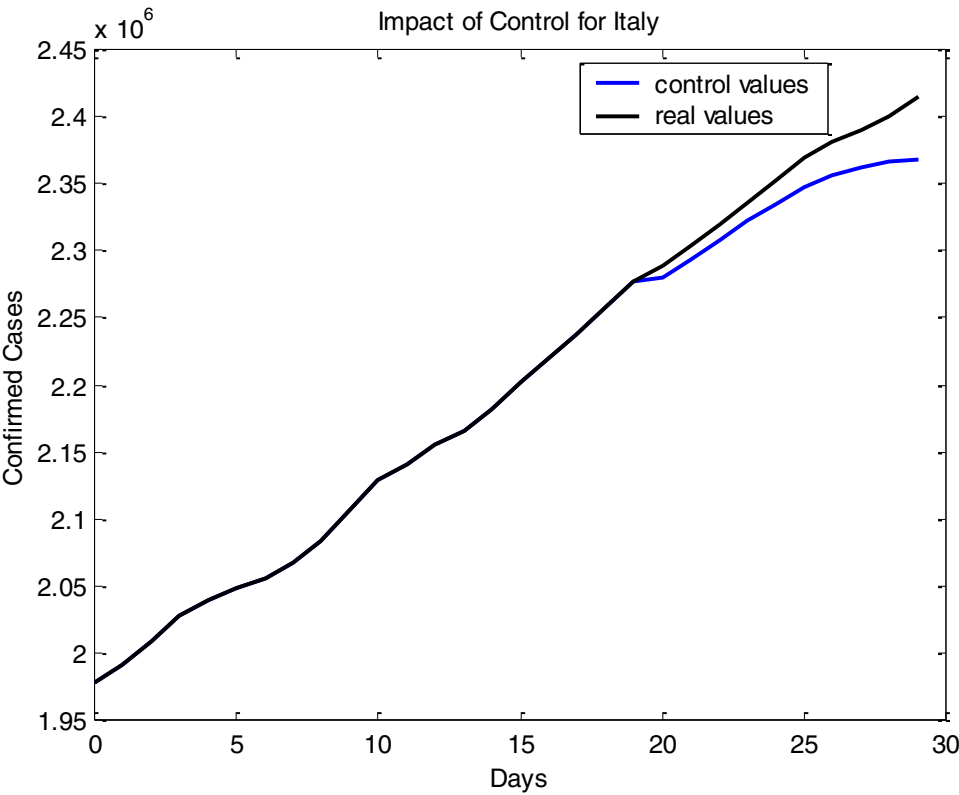


Fig. 19. Plot of real confirmed cases (black) and cases under the impact of the controller (blue) for Italy.

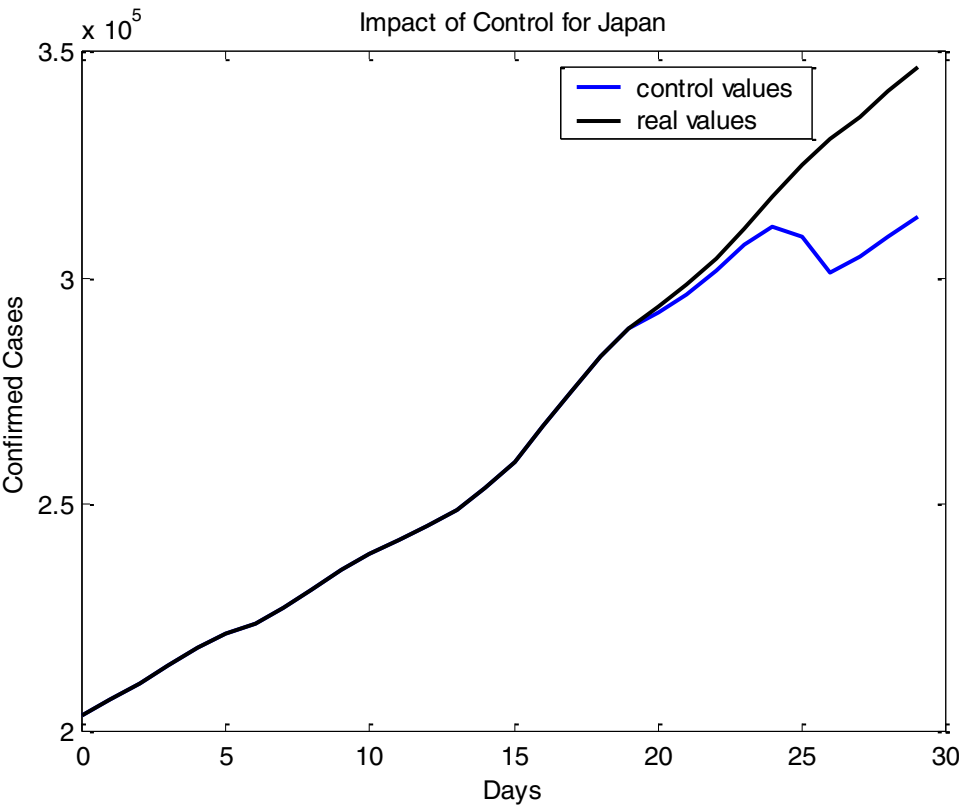


Fig. 20. Plot of real confirmed cases (black) and cases under the impact of the controller (blue) for Japan.

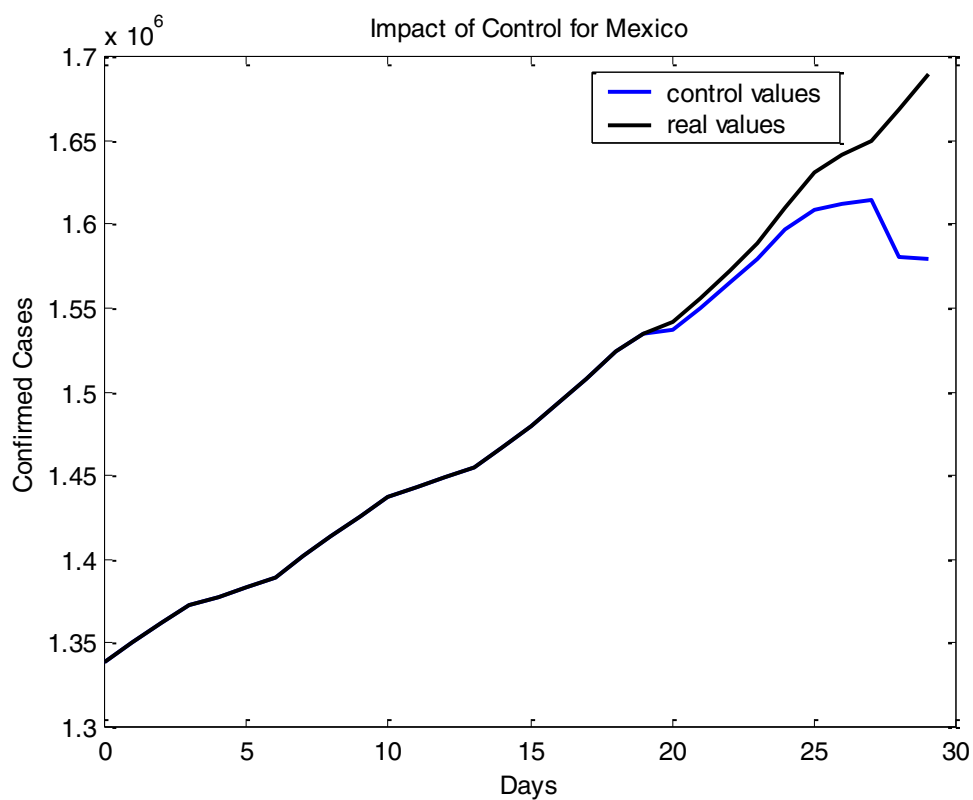


Fig. 21. Plot of real confirmed cases (black) and cases under the impact of the controller (blue) for Mexico.

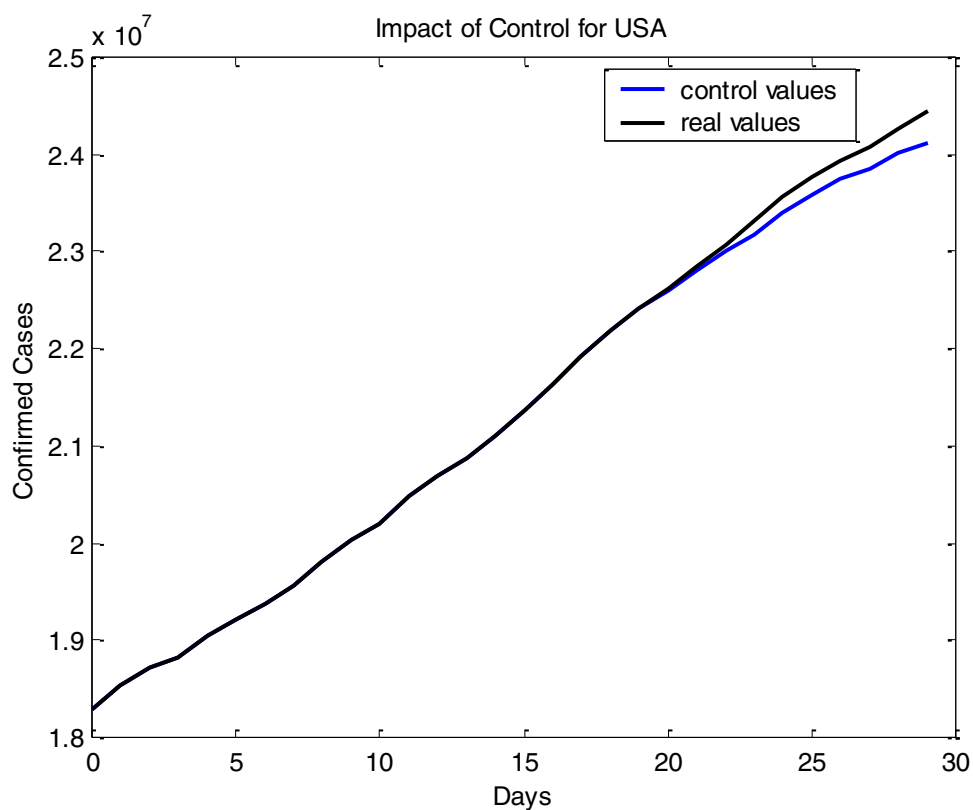


Fig. 22. Plot of real confirmed cases (black) and cases under the impact of the controller (blue) for USA.

fuzzy rules and membership functions was done by trial and error, in combination with knowledge about the problem, but we believe that the application of an optimization method can improve the results, and we will consider this task in our future work. We believe that metaheuristics coming from computational intelligence could be easily used to optimize the fuzzy model for control and further improve the results.

7. Conclusion

In this article an approach for controlling the COVID-19 pandemics using an appropriate mixture of fractal theoretical constructs and fuzzy logic concepts has been outlined. An approximation of the fractal dimension based on the box counting algorithm was utilized to estimate the complexity of the dynamics of the countries. Fuzzy Logic was utilized to model the intrinsic uncertainty in the decision-making involved for achieving the control goal. The method is comprised by a fuzzy model, formed with fuzzy rules, that utilizes the fractal dimensions as input values and calculates as outputs the control actions to be performed on the countries based on the COVID-19 data at each point in time. The main contribution of this article is the hybrid method mixing in an appropriate fashion the fractal dimension and fuzzy logic theoretical constructs for achieving an efficient and relatively simple control of the COVID-19 pandemics. Publicly available data sets of 7 countries have been used to experiment and build the fuzzy system with fractal dimensions as fuzzy variables, and a period of time was used to validate the approach by measuring the effect of the control in reducing the number of confirmed cases. We are imagining as future research work the utilization of the proposed method to similar control problems, as well as augmenting the order of fuzzy logic to type-2, which theoretically speaking should provide a better representation of the uncertainty in the decision-making process that is required for achieving the control goal. In addition, we are envisioning studying the relations of this article to other recent relevant works on other facets of the COVID-19 problem, like the ones presented in [42,43], or in forecasting [44,45]. Finally, we also envision combining our approach with new neural network models to control the Pandemic both on the spatial and temporal dimensions, like in [46,47].

Credit Author Statement

Oscar Castillo did his work on the fuzzy fractal model for control, and then validated the implementation and the results. Patricia Melin proposed the method and the experiments that were performed, then implemented the proposed method, and contributed to the simulations. Both authors documented the results and prepared the manuscript, as well as worked on enhancing quality of writing

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Zadeh LA. "Fuzzy Sets". *Inform Control* 1965;8:338–53.
- [2] Zadeh LA. The Concept of a Linguistic Variable and its Application to Approximate Reasoning. *Inform Sci* 1975;8:43–80.
- [3] Mendel JM. *Uncertain Rule-Based Fuzzy Systems. Introduction and New Directions*, 2nd Edition, 2nd ed. Springer; 2017.
- [4] Castillo O, Castro JR, Melin P, Rodriguez-Diaz A. Application of interval type-2 fuzzy neural networks in non-linear identification and time series prediction. *Soft Comput* 2014;18(6):1213–24.
- [5] Castillo O. *Type-2 Fuzzy Logic in Intelligent Control Applications*. Springer; 2012. Berlin, Germany.
- [6] Ontiveros-Robles E, Melin P, Castillo O. Comparative analysis of noise robustness of type 2 fuzzy logic controllers. *Kybernetika* 2018;54:175–201.
- [7] Mandelbrot B. "The Fractal Geometry of Nature", W.H. Freeman and Company; 1987.
- [8] Castillo O, Melin P. Developing a New Method for the Identification of Microorganisms for the Food Industry using the Fractal Dimension. *J Fractals* 1994;2(3):457–60.
- [9] Melin P, Castillo O. "An Adaptive Model-Based Neuro-Fuzzy-Fractal Controller for Biochemical Reactors in the Food Industry", *Proceedings of IJCNN'98*, 1. IEEE Computer Society Press; 1998. p. 106–11. Alaska, USA.
- [10] Castillo O, Melin P. A New Method for Fuzzy Estimation of the Fractal Dimension and its Applications to Time Series Analysis and Pattern Recognition. *Proceedings of NAFIPS'2000*; 2000. Atlanta, GA, USA p. 451–5.
- [11] The Humanitarian Data Exchange (HDX). [Online]. Available: <https://data.humdata.org/dataset/novel-coronavirus-2019-ncov-cases>. [Accessed 31 03 2020].
- [12] Shereen MA, Khan S, Kazmi A, Bashir N, Siddique R. COVID-19 infection: origin, transmission, and characteristics of human coronaviruses. *J Adv Res* 2020;24:91–8.
- [13] Sohrabi C, Alsafi Z, O'Neill N, Khan M, Kerwan A, Al-Jabir A, Agha R. World Health Organization declares global emergency: A review of the 2019 novel coronavirus (COVID-19). *Int J Surg* 2020;76:71–6.
- [14] Apostolopoulos ID, Bessiana T. COVID-19: Automatic detection from X-Ray images utilizing Transfer Learning with Convolutional Neural Networks; 2020. arXiv preprint arXiv:2003.11617.
- [15] Sarkodie SA, Owusu PA. Investigating the Cases of Novel Coronavirus Disease (COVID-19) in China Using Dynamic Statistical Techniques; 2020. Available at SSRN 3559456.
- [16] Beck BR, Shin B, Choi Y, Park S, Kang K. Predicting commercially available antiviral drugs that may act on the novel coronavirus (SARS-CoV-2) through a drug-target interaction deep learning model. *Comput Struct Biotechnol J* 2020;18:784–90.
- [17] Zhong L, Mu L, Li J, Wang J, Yin Z, Liu D. Early Prediction of the 2019 Novel Coronavirus Outbreak in the Mainland China based on Simple Mathematical Model. *IEEE Access* 2020;8:51761–9.
- [18] Kamel Boulos MN, Geraghty EM. Geographical tracking and mapping of coronavirus disease COVID-19/severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) epidemic and associated events around the world: how 21st century GIS technologies are supporting the global fight against outbreaks and epidemics. *Int J Health Geogr* 2020;19(8). doi:10.1186/s12942-020-00202-8.
- [19] Gao P, Zhang H, Wu Z, Wang J. Visualising the expansion and spread of coronavirus disease 2019 by cartograms. *Environ Plan A* 2020. doi:10.1177/0308518X20910162.
- [20] Rao ASRS, Vazquez JA. Identification of COVID-19 can be quicker through artificial intelligence framework using a mobile phone-based survey in the populations when Cities/Towns are under quarantine. *Infect Control Hosp Epidemiol* 2020. doi:10.1017/ice.2020.61.
- [21] Santosh KC. AI-driven tools for coronavirus outbreak: Need of active learning and cross-population Train/Test models on Multitudinal/Multimodal data. *J Med Syst* 2020;44(5). doi:10.1007/s10916-020-01562-1.
- [22] Robson B. Preliminary bioinformatics studies on the design of a synthetic vaccine and a preventative peptidomimetic antagonist against the SARS-CoV-2 (2019-nCoV, COVID-19) coronavirus. *Comput Biol Med* 2020;119:1–19.
- [23] Fanelli D, Piazza F. Analysis and Forecast of COVID-19 spreading in China. *Italy and France. Chaos Solitons Fractals* 2020;134:1–5.
- [24] Castillo O, Melin P. "A new Fuzzy Inference System for Reasoning with Multiple Differential Equations for Modelling Complex Dynamical Systems". *IOS Press*; 1999. p. 224–9. Vienna Austria.
- [25] Castillo, O. and Melin, P. (1998). "A new Fuzzy-Fractal-Genetic Method for Automated Mathematical Modelling and Simulation of Robotic Dynamic Systems", *Proceedings of FUZZ'98*, IEEE Press, Alaska, USA, Vol. 2, pp. 1182–1187.
- [26] Haidegger T, Kovacs L, Precup R-E, Preitl S, Benyo B, Benyo Z. Cascade control for telerobotic systems serving space medicine, Volumes 44; 2011. p. 3759–64. (IFAC Proceedings).
- [27] Precup R-E, Tomescu M-L. Stable fuzzy logic control of a general class of chaotic systems. *Neural Comput Appl* 2015;26(3):541–50.
- [28] Zhang X, Lin W. A K-filter-based adaptive control for nonlinear systems with unknown parameters in state and output equations. *Automatica* 2019;105:186–97.
- [29] Das A, Dhar A, Goyal S, Kundu A, Pandey S. COVID-19: Analytic results for a modified SEIR model and comparison of different intervention strategies. *Chaos Solitons Fractals* 2021;144:1–13 110595.
- [30] Nabi KN, Kumar P, Erturk VS. Projections and fractional dynamics of COVID-19 with optimal control strategies. *Chaos Solitons Fractals* 2021;145:1–16 110689.
- [31] Kouidere A, Youssoufi LE, Ferjouchia H, Balatif O, Rachik M. Optimal Control of Mathematical modeling of the spread of the COVID-19 pandemic with highlighting the negative impact of quarantine on diabetics people with Cost-effectiveness. *Chaos Solitons Fractals* 2021;145:1–12 110777.
- [32] Shakhany MQ, Salimifard K. Predicting the dynamical behavior of COVID-19 epidemic and the effect of control strategies, 146; 2021. p. 1–11. 110823.
- [33] Asamoah JK, Jin Z, Sun G-Q, Seidu B, Yankson E, Abidemi A, Oduro FT, Moore SE, Okyere E. Sensitivity assessment and optimal economic evaluation of a new COVID-19 compartmental epidemic model with control interventions, 146; 2021. p. 1–19. 110885.

- [34] Chakraborty T, Ghosh I. Real-time forecasts and risk assessment of novel coronavirus (COVID-19) cases: A data-driven analysis. *Chaos, Solitons Fractals* 2020;135:109850.
- [35] Bezdek JC. *Pattern Recognition with Fuzzy Objective Function Algorithms*. Plenum Press; 1981.
- [36] Yager R, Filev D. *Generation of Fuzzy Rules by Mountain Clustering*. *Intell Fuzzy Syst* 1994;2(3):209–19.
- [37] Sugeno M, Kang GT. *Structure Identification of Fuzzy Model*. *Fuzzy Sets Syst* 1988;28:15–33.
- [38] Jang JR, Sun CT, Mizutani E. *Neuro-Fuzzy and Soft Computing*. Prentice Hall; 1997.
- [39] Contreras S. A multi-group SEIRA model for the spread of COVID-19 among heterogeneous populations. *Chaos, Solitons Fractals* 2020;136:1099325.
- [40] Brauer F, Castillo-Chavez C. *Mathematical Models in Population Biology and Epidemiology*. Springer; 2012. New Yorkfirst ed.
- [41] Ridenhour B, Kowalik JM, Shay DK. «El número reproductivo básico (R0): consideraciones para su aplicación en la salud pública*,». *Am J Public Health* 2018;108:S455–65. doi:10.2105/AJPH.2013.301704s.
- [42] Crokidakis N. COVID-19 spreading in Rio de Janeiro, Brazil: Do the policies of social isolation really work? *Chaos, Solitons Fractals* 2020;136:109930.
- [43] Adbo MS. On a comprehensive model of the novel coronavirus (COVID-19) under Mittag-Leffler derivative. *Chaos, Solitons Fractals* 2020;135:109867.
- [44] Boccaletti S. Modeling and forecasting of epidemic spreading: The case of Covid-19 and beyond. *Chaos, Solitons Fractals* 2020;135:109794.
- [45] Mandal et al M. A model based study on the dynamics of COVID-19: Prediction and control. *Chaos, Solitons Fractals* 2020;136:109889.
- [46] Melin P, Monica JC, Sanchez D, Castillo O. Analysis of Spatial Spread Relationships of Coronavirus (COVID-19) Pandemic in the World using Self Organizing Maps. *Chaos Solitons Fractals* 2020;138:1–7 1099172020.
- [47] Melin P, Monica JC, Sanchez D, Castillo O. Multiple Ensemble Neural Network Models with Fuzzy Response Aggregation for Predicting COVID-19 Time Series: The Case of Mexico. *Healthcare* 2020;8:181.